

Demo 7

Mapping Color Patterns onto a Phylogeny

Do closely related reef fishes look alike?



EEB 187: Ecology and Evolution of Color | Week 8 Demo

The source study

Frédérich et al. 2026: Rapid and repeated evolution of pigmentation patterns in reef fishes

Core question Do reef-fish color patterns accumulate gradually with geography, or diversify rapidly as lineages split?

918

species

1,799

fish images

6

reef-fish families

30

motif categories

3

body regions

Their pipeline

photos -> motif presence/absence matrix

matrix -> Gower distances -> PCoA pattern space

pattern space + phylogeny -> signal, disparity, and transition timing

Today's demo uses the Acanthuridae slice so students can see the logic without rebuilding the whole paper.

What did they measure?

Not spectra: named visual motifs scored from adult fish images



one color



horizontal stripe



vertical stripes



oblique stripes



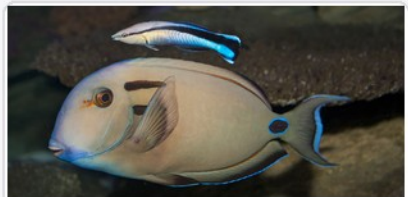
blotch



dots



eyespot



marbling



saddle

A motif is a visible pattern element

stripe orientation: horizontal, vertical, oblique

localized elements: dots, blotches, eyespots, saddles

complex texture: marbling or honeycomb

special structures: mouth, operculum, caudal fin, scalpel

Why the SI matters

Additional file 3 defines each motif with example images, so the coding scheme is teachable and reproducible.

What does the raw data look like?

Data S1 is a presence/absence matrix: species in rows, geography and motifs in columns

species	CIP	WI	CP	A	TEP	H 1 h-stripe	H 2+ h-stripe	H marbling	Tr 1 h-stripe	Tr 2+ h-stripe	Tr 1 dot	Tr marbling	Ta 2+ dots	Ta colored sc
Acanthurus achilles	1	0	1	0	1	0	0	0	0	1	0	0	0	0
Acanthurus albimento	1	0	0	0	0	0	0	0	0	0	0	1	0	1
Acanthurus albipectoralis	1	0	0	0	0	0	0	1	0	0	0	0	0	0
Acanthurus auranticavus	1	1	0	0	0	0	0	0	0	0	0	1	0	0
Acanthurus bahianus	0	1	0	1	0	0	0	0	0	0	0	0	1	1
Acanthurus bariene	1	1	1	0	0	0	0	1	0	0	0	1	1	1
Acanthurus blochii	1	1	1	0	0	0	0	0	0	0	0	0	1	1

Yellow columns

biogeographic regions coded
0/1

Orange columns

motifs repeated for head, trunk,
tail

The important move

a photograph becomes
analyzable trait data

This is very different from measuring color spectra or image patch contrast with pavo.

Motif coding is a different kind of color-

Useful, scalable, and deliberately coarse

pattern data

Last week: pavo-style image/color analysis

continuous measurements

patch geometry and boundaries

chromatic and achromatic contrast

asks: how much color difference is visible?

This week: Frédéric motif coding

categorical presence/absence

body-region vocabulary: head, trunk, tail

works across hundreds of species

asks: which pattern elements are present?

Tradeoff: motif coding loses fine color and contrast information, but gains a broad comparative dataset.

The published data package

Figshare contains the raw matrix, transformed coordinates, geography, tree, and R code

Additional_file_1_Data.xlsx

raw motif + region 0/1 matrix for six families

coord_Acanthuridae.txt

PCoA coordinates: species positions in pattern space

coord_Acanthuridae_ar.txt

same coordinates, duplicated for species in multiple regions

grouping_Acanthuridae.txt

region membership table for diversity indices

tree_Acanthuridae.tree

time-calibrated phylogeny for the demo family

Color_patterns_evolution_Code.R

analysis code used to produce paper results

Our demo is the teaching version of this workflow: start with Acanthuridae, make simple trait summaries, then map them on a tree.

More species, more motif space

Figure 1 links species richness to pigmentation pattern richness across regions

Frédérich et al. BMC Biology (2026) 24:46

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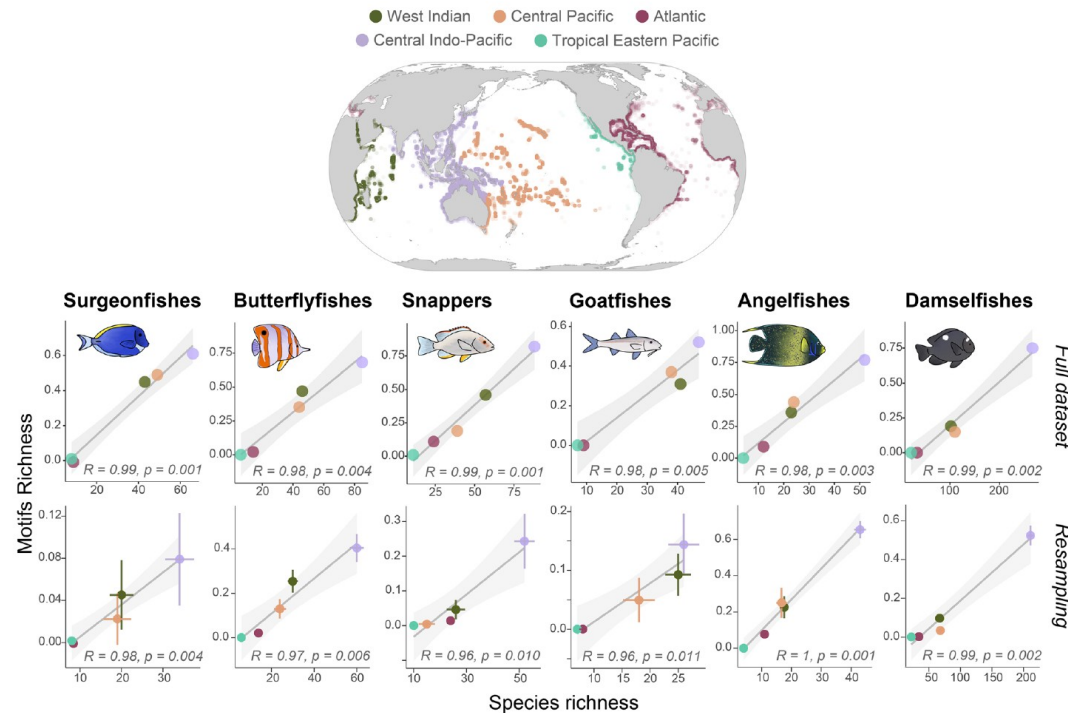


Fig. 1 Relationships between motif richness and species richness from the five main biogeographic regions. The richness of pigmentation patterns (i.e., motif richness) is directly proportional to the number of species present in each region. Tests of correlation are all significant and Pearson's

How to read it

each point = one biogeographic region in one family

x-axis = number of species in the region

y-axis = motif richness, measured in pattern space

result: richer assemblages occupy more motif space

Teaching hook

Motif diversity scales with species richness, not just with region identity.

Rapid but bounded evolution

Figure 2 asks whether pattern disparity is higher within recent subclades than expected

high level of homoplasy observed in animal coloration, diversification reflects highly rapid evolution within a

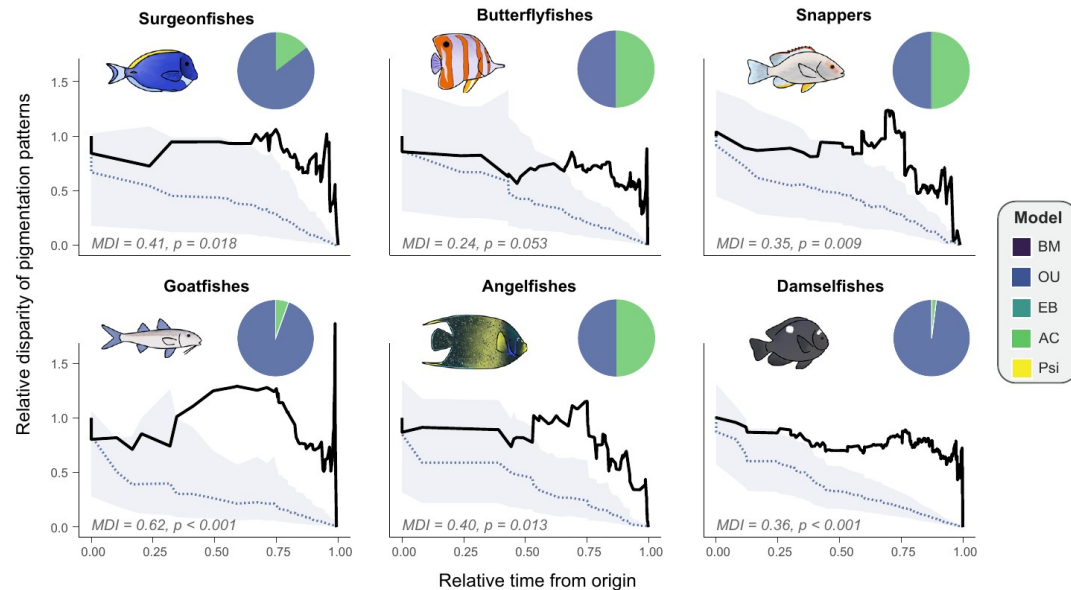


Fig. 2 Disparity through time (DTT) plots illustrating the evolution of subclade disparity across the phylogeny of each fish family. Higher relative disparity values (y-axis) correspond to a greater average volume of pigmentation pattern space occupied by subclades relative to the disparity of the entire family. The solid line represents the observed disparity calculated for each clade, while dashed line is median expected disparity under a null Brownian motion model based on simulations. Across all fish families, subclade disparity was significantly higher than expected under BM, with sharp increases of within subclade disparity toward the present, suggesting the potential for high motif convergence between subclades. Inset pie charts display the distribution of the Akaike information criterion weights (AICw) from trait evolutionary model (BM, OU, EB, AC, and psi) comparisons. Generally, support for BM, EB, and psi models was too low to be discernible on pie charts. Plots reflect model comparisons for PC1. Plots of PCs 2–3 are provided in Additional file 2: Fig. S2 and Additional file 2: Fig. S3

Translation for today

DTT = disparity through time

green line above null = close relatives are more different than Brownian expectation

AC and OU-style models fit many axes

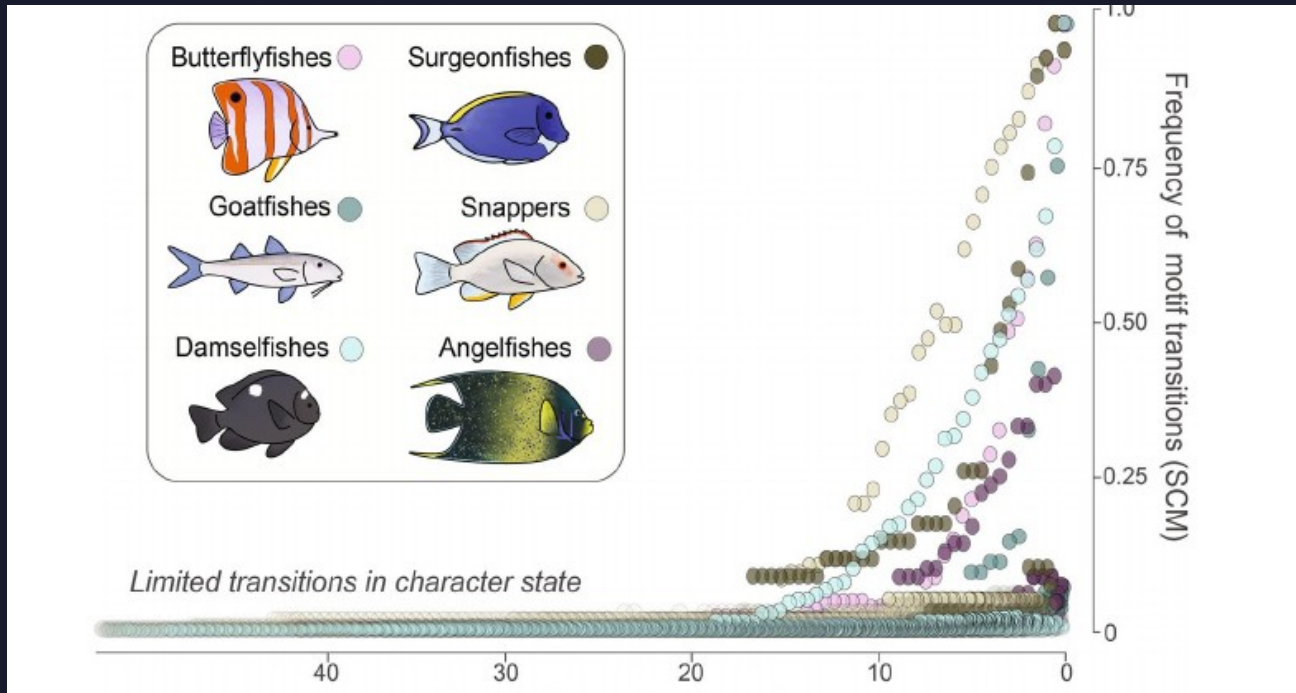
interpretation: patterns change fast, but within a bounded motif space

Useful number

Motif half-life was about 1.5 Ma, far faster than many fish morphological traits.

Motif transitions are recent

Figure 3 uses stochastic character maps to ask when motif gains and losses occurred



What the dots mean

each circle counts inferred motif transitions at a time point

low values deep in time = little inferred motif turnover

many transitions in the last 5-10 Ma

supports recent, repeated pattern diversification

Connection to demo

When you map one trait on a tree, you are doing the first step toward this kind of transition history.

What did Frédéricich et al. conclude?

A compact version for today's lab

Species richness matters

Regions with more species contain more motif diversity.

Signal is low but significant

Relatedness matters, but close relatives often diverge strongly.

Evolution is rapid

Pattern shifts are concentrated near the tips of the tree.

Evolution is bounded

Lineages repeatedly explore a limited vocabulary of motifs.

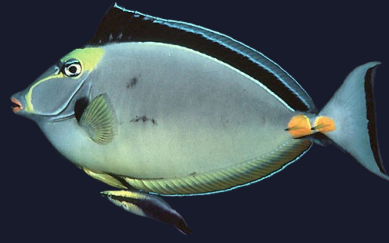
Speciation is central

Pattern disparity is high within recent subclades, consistent with repeated divergence among close relatives.

Today: use one family to make those ideas concrete.

Today's slice of the paper

Acanthuridae: surgeonfishes, tangs, and unicornfishes



Published inputs

tree + motif matrix

Your analysis

complexity + one motif

Your interpretation

does pattern follow the
tree?

The demo is intentionally smaller than the paper, but the reasoning is the same.

Today's big question

Do closely related fish species have similar color patterns?

Or do selection, communication, and convergence repeatedly overwrite the family tree?

If signal is strong

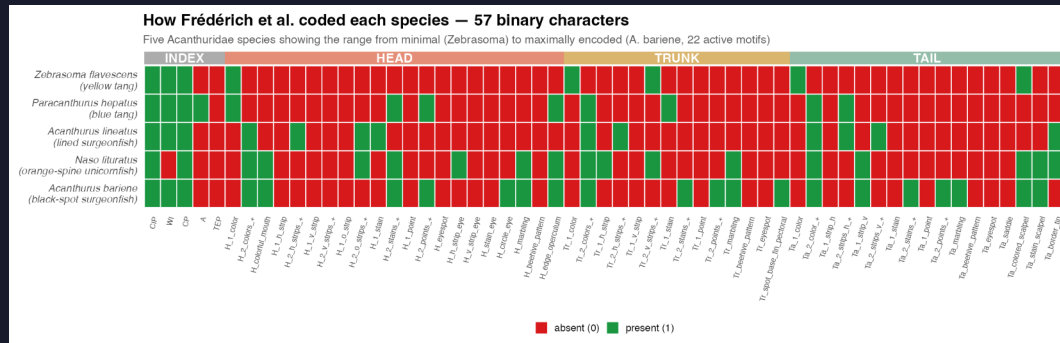
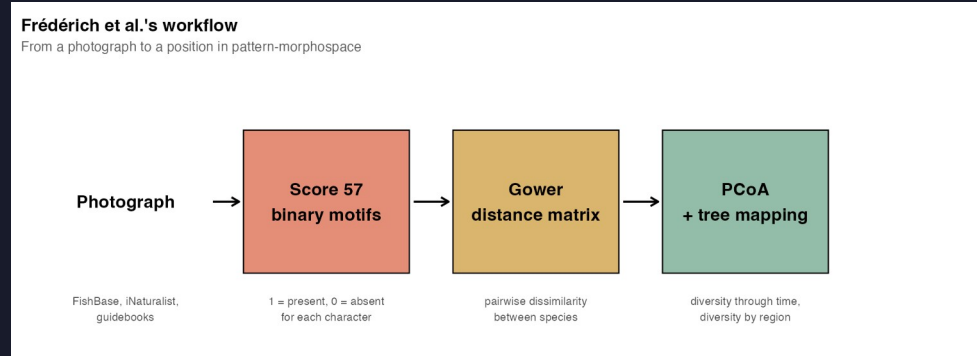
close relatives look alike

If signal is weak

patterns evolve fast or converge

We answer with a tree, a color-pattern matrix, and two statistics.

From photograph to comparative data



The key move

Species are not compared by vibes.

Each fish is scored for visible motifs across body regions.

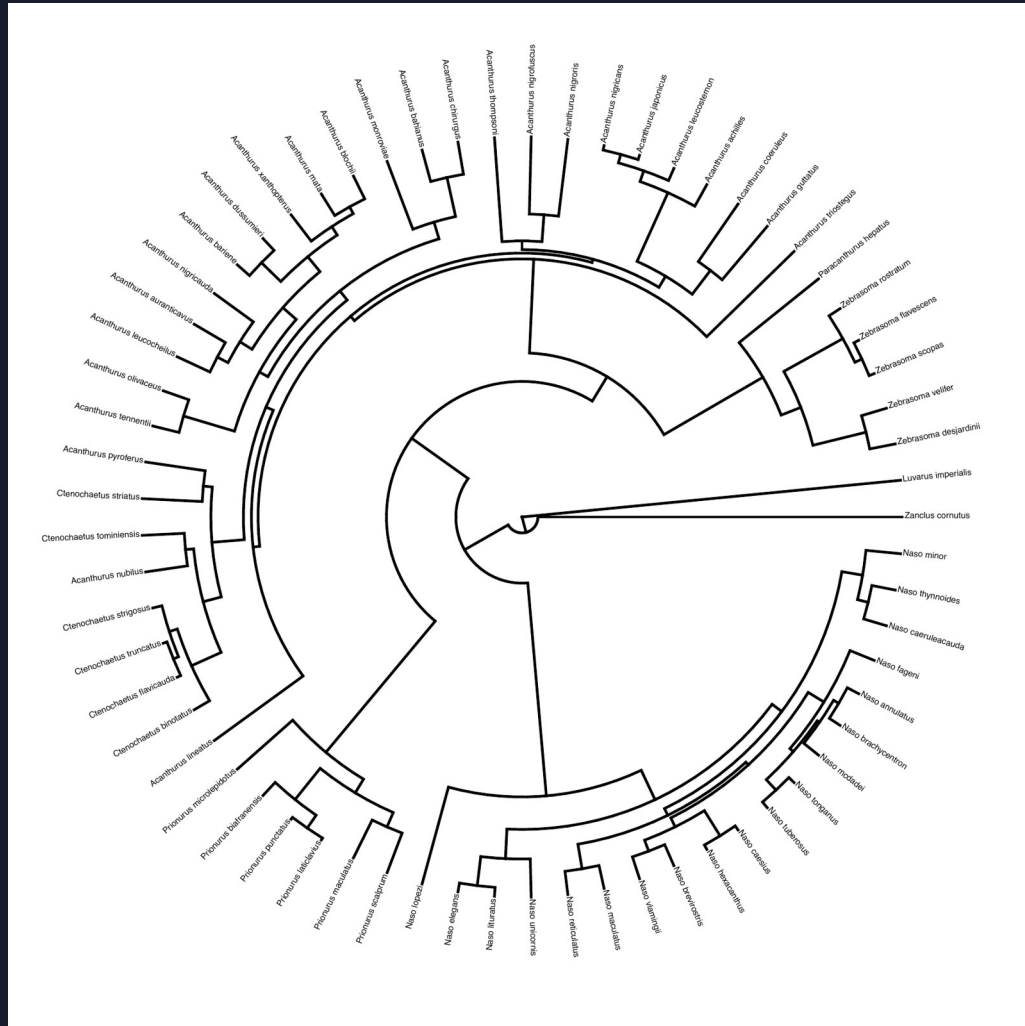
Every species becomes a row of 0s and 1s.

Then we can ask whether pattern similarity follows the tree.

Important limitation

This demo scores pattern motifs, not full color spectra. That is why your later pavo work matters.

What is a phylogeny?



A family tree for species

tips = living species

internal nodes = common ancestors

branch lengths = time or amount of evolution

nearby tips share more history than distant tips

Today's tree

Acanthuridae: surgeonfishes, tangs, unicornfishes

Phylogenetic signal: does the tree predict the trait?

Strong signal



relatives share similar values

Weak signal

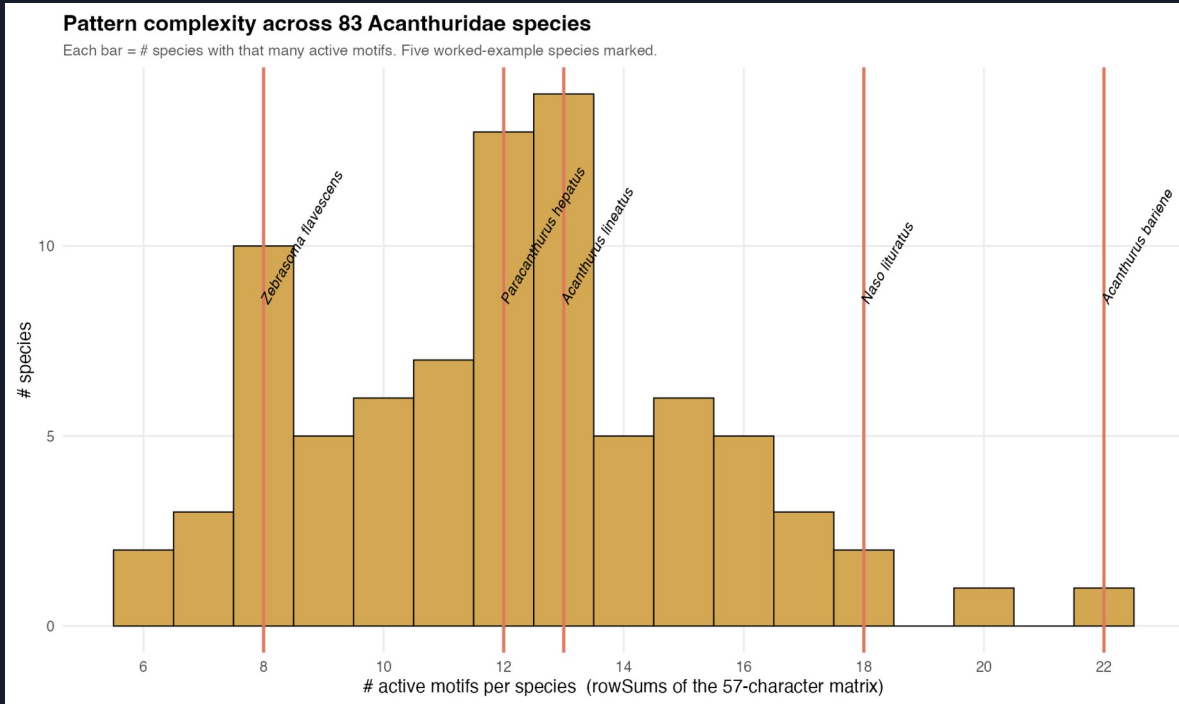


close relatives can look different

Demo question: where do surgeonfish color patterns fall?

Pattern complexity

One number per species



How we summarize 57 motif columns

count how many pattern motifs are present

simple species have few active motif columns

complex species have many active motif columns

then map that number onto the tree

Interpret carefully

Complexity is useful, but it does not tell us whether motifs serve the same function.

Painting complexity onto the tree

Painting complexity onto the tree

[illegible]

- hot branches: complex patterns
- cool branches: simpler patterns
- sudden color shifts: candidate evolutionary changes
- clusters: possible phylogenetic conservation

Internal branch colors are inferred ancestral values, not observed data.

Two ways to quantify phylogenetic signal

Blomberg's K

Compares observed trait similarity to a Brownian-motion expectation.

$K = 0$

random on tree

$K = 1$

Brownian

$K > 1$

extra conserved

Pagel's λ

Asks how much the tree needs to be transformed to explain the trait.

$\lambda = 0$

tree irrelevant

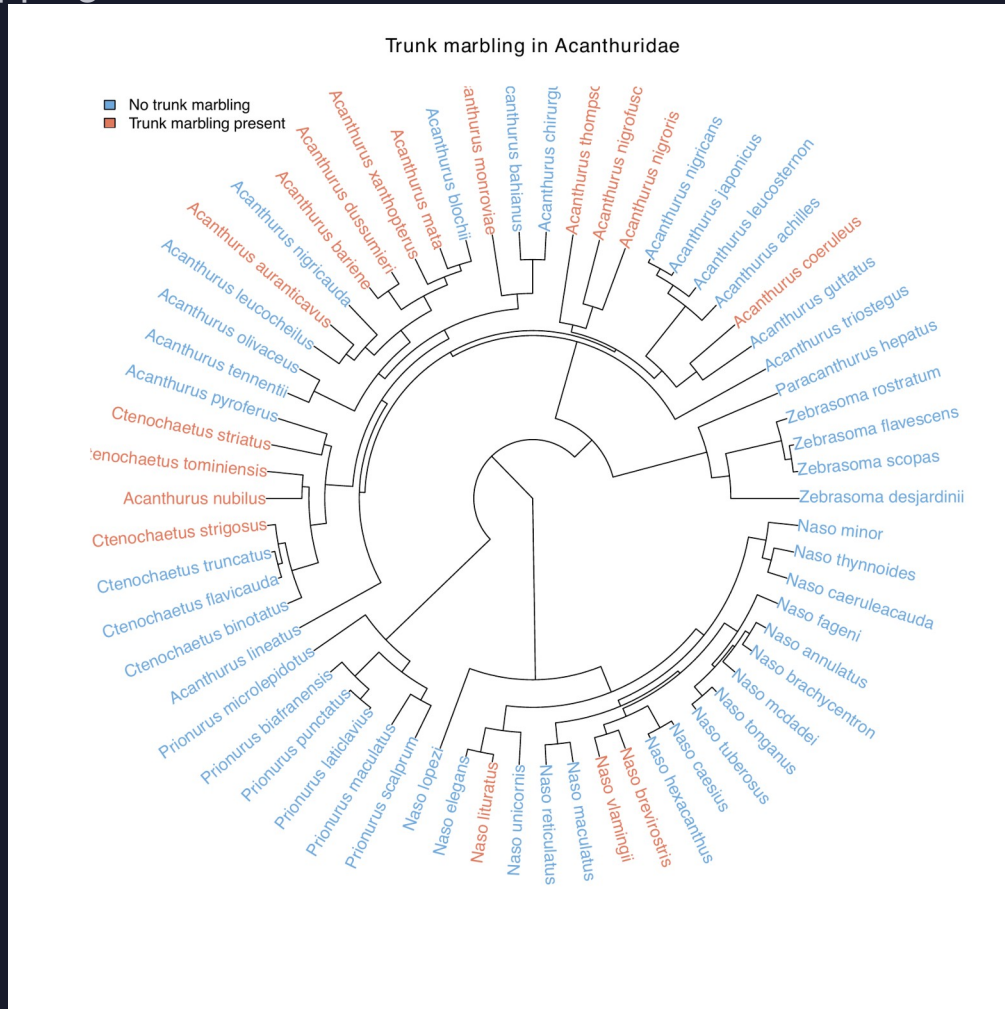
$\lambda = 1$

tree fits

For our complexity score: K is modest but significant; that is the interesting middle ground.

Discrete traits

Mapping one motif at a time



Example: trunk marbling

binary: present or absent

tip colors show the observed states

clustering suggests conservation

scattered tips suggest repeated gains/losses

Discussion prompt

Could marbling be a Turing-like labyrinth pattern? What evidence would you want?

Your group becomes the data collector

83

TOTAL SPECIES

0

REMOVED

0

ALPHABETICALLY

863

TOTAL RANGES

0.0% complete

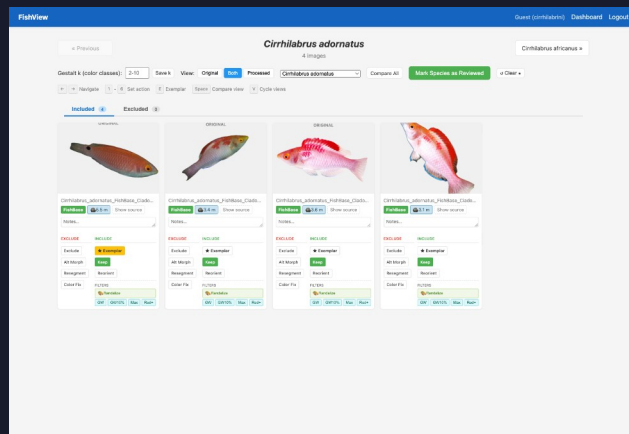
Filter species...

☒ Only species in the tree (55)
 ☐ Exact PNC 2 Result (1) in example 1 tree

All statuses ▾

Jump to species...

SPECIES	IMAGES	REMOVED	EXCLUDED	EXEMPLAR	K	STATUS
<i>Cirrhlabrus aeneus</i>	4	0	0	1	-	Not Started
<i>Cirrhlabrus alticus</i>	8	0	0	-	-	Not Started
<i>Cirrhlabrus asoteryx</i>	1	0	0	-	-	Not Started
<i>Cirrhlabrus aurantimaculatus</i>	2	0	0	-	-	Not Started
<i>Cirrhlabrus holbrooki</i>	17	0	0	-	-	Not Started
<i>Cirrhlabrus kuetzingii</i>	5	0	0	-	-	Not Started
<i>Cirrhlabrus leucophaea</i>	12	0	0	-	-	Not Started
<i>Cirrhlabrus macleayi</i>	8	0	0	-	-	Not Started
<i>Cirrhlabrus nigrifrons</i>	3	0	0	-	-	Not Started
<i>Cirrhlabrus novaeae</i>	1	0	0	-	-	Not Started
<i>Cirrhlabrus ocellatus</i>	5	0	0	-	-	Not Started
<i>Cirrhlabrus obscurus</i>	3	0	0	-	-	Not Started
<i>Cirrhlabrus opakei</i>	14	0	0	-	-	Not Started
<i>Cirrhlabrus cyanoguttatus</i>	12	0	0	-	-	Not Started
<i>Cirrhlabrus cyanopterus</i>	23	0	0	-	-	Not Started



FishView workflow

log in with your group code

filter to species that are in your tree

choose three species from three genera

star one exemplar image per species

export a bundle for the R helper

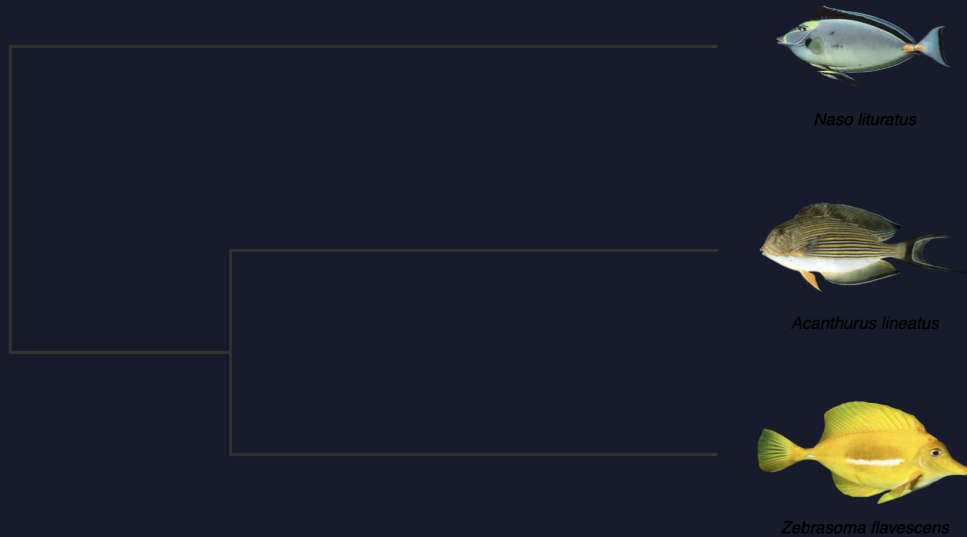
The logic is the same

Part 1 uses published coded traits. Part 2 uses your chosen fish images on your group's tree.

Final figure

A tree with evidence attached

Part 2: acanthuridae – your three exemplars on the tree



What this lets you ask

Do close relatives look similar?

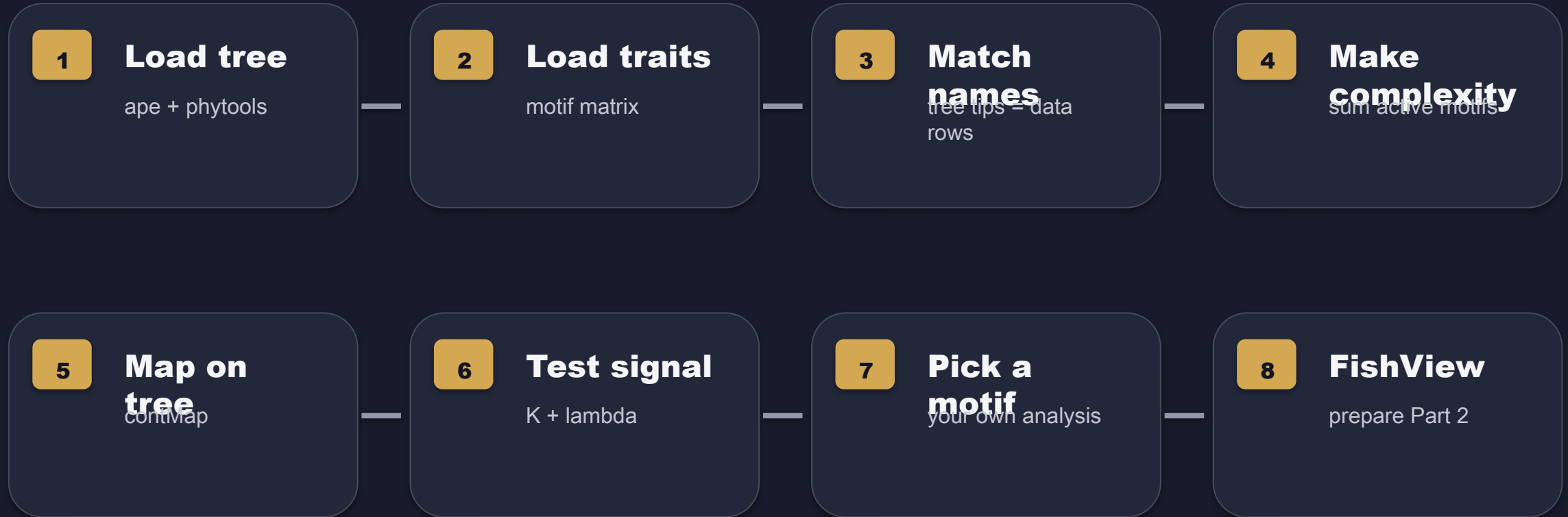
Are pattern differences clustered by genus?

Are color changes tied to ecology, sex, or life stage?

What would pavo metrics add that motif coding misses?

This is the bridge to your final project.

What you will do today



Deliverable: contMap screenshot + K/lambda interpretations + one-paragraph biological answer.

What you will turn in

1. Screenshot of your contMap for pattern complexity
2. Blomberg's K and Pagel's lambda, each interpreted in one sentence
3. Your chosen trait: result + whether it shows signal
4. One paragraph answering: is color pattern diversity phylogenetically conserved in surgeonfishes?

The goal is not one magic p-value.

The goal is to connect a statistical result to an evolutionary story.

Let's go.